Technical Requirement Document

Group No. XX
Project Title
BS COMPUTER SCIENCE
BATCH 2022F
Supervisor' Name
Supervisor's Designation
SSUET

Submitted by

2022F-BSCS-XXX	Student Name
2022F-BSCS-XXX	Student Name
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In Partial Fulfillment
Of the Requirements for the degree
Bachelor of Science in Computer Science

Department of Computer Science & Information Technology
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November 2025

DECLARATION

We hereby declare that this project report entitled **"TITLE"** submitted to the **"Department of Computer Science and Information Technology",** is a record of an original work done by us under the guidance of Supervisor "**NAME**" and that no part has been plagiarized without citations. Also, this project work is submitted in the partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science.

Name : Student Name	Signature: ______________________
Name : Student Name	Signature: ______________________
Name : Student Name	Signature: ______________________
Name : Student Name	Signature: ______________________

Supervisor:	**Signature:**
Supervisor Name Designation	______________________
Date	______________________

ABSTRACT

Include a brief summary of the problem statement, challenges, proposed solution, approaches, scope, and comparison with existing systems/evaluation methods, conclusion and future directions. Recommend length is 1 page maximum. Abstract must be self-explanatory and should not include any references or short hand notations in the abstract.

Table of Content

List of Figures

List of Tables

Chapter 1

Introduction

1.1 Background of Study

[In first paragraph, explain the area or domain from which you have chosen your project. In Second paragraph, explain specifically, about the project.]

1.3 Problem Statement

[State clear problem(s) to be solve by your project.]

1.2 Project Objectives

[Describes the desired results of a project]

1.4 Scope of Study

[Describes what the product will accomplish.]

Chapter 2

Literature Review

2.1 Existing Systems

< Discuss at least 3 to 4 most relevant projects with title, brief description and system features with references show as follows. >

2.1.1 Title: <Project Title> [1]

Description:

System Features:

2.1.2 Title: <Project Title> [2]

Description:

System Features:

2.2 Proposed System:

2.2 Comparative Analysis:

<Enlist the unique features of these 3 to 4 projects and chose 5 to 6 to be included in your project. Moreover, present comparative analysis of your project with these existing by features.>

Chapter 3

Requirement Specifications

3.1 External Interface Requirements

3.1.1 User Interfaces

<Describe the logical characteristics of each interface between the software product and the users>

3.1.2 Hardware Interfaces

<Describe the logical and physical characteristics of each interface between the software product and the hardware components of the system.>

3.1.3 Software Interfaces

<Describe the connections between this product and other specific software components (name and version), including databases, operating systems, tools, libraries, and integrated commercial components. Identify the data items or messages coming into the system and going out and describe the purpose of each. Describe the services needed and the nature of communications.>

3.1.4 Communications Interfaces

<Describe the requirements associated with any communications functions required by this product, including e-mail, web browser, network server communications protocols, electronic forms, and so on. Define any pertinent message formatting. Identify any communication standards that will be used, such as FTP or HTTP. Specify any communication security or encryption issues, data transfer rates, and synchronization mechanisms.>

3.2 Functional Requirements

<Specify Function requirements. Each requirement should be uniquely identified with a sequence number or a meaningful tag of some kind as shown in ***Table.1***.>

TABLE 1: Put Caption against Each Table Here

ID	Requirement
REQ-1	
REQ-2	
REQ-3	
REQ-4	
REQ-5	

3.3 Other Nonfunctional Requirements

3.3.1 Performance Requirements

< Specify Performance Requirement >

3.3.2 Safety Requirements

< Specify Safety Requirement >

3.3.3 Security Requirements

< Specify Security Requirement >

3.3.4 Software Quality Attributes

<Specify any additional quality characteristics for the product that will be important to either the customers or the developers. Some to consider are: adaptability, availability, correctness, flexibility, interoperability, maintainability, portability, reliability, reusability, robustness, testability, and usability. Write these to be specific, quantitative, and verifiable when possible. At the least, clarify the relative preferences for various attributes, such as ease of use over ease of learning.>

3.4 Cost Estimation

S.No	Project Expenditure	Cost in Rupees
1	**Software Tools and API** • Name of APIs • Name of Software • Name of Framework • Name of Software Tools • Name of Server	 **Sub Total**
2	**Hardware Cost** • Equipment 1 • Equipment 2 • Equipment 3	 **Sub Total**
3	**Networking Cost** • Equipment 1 • Equipment 2 • Equipment 3	 **Sub Total**

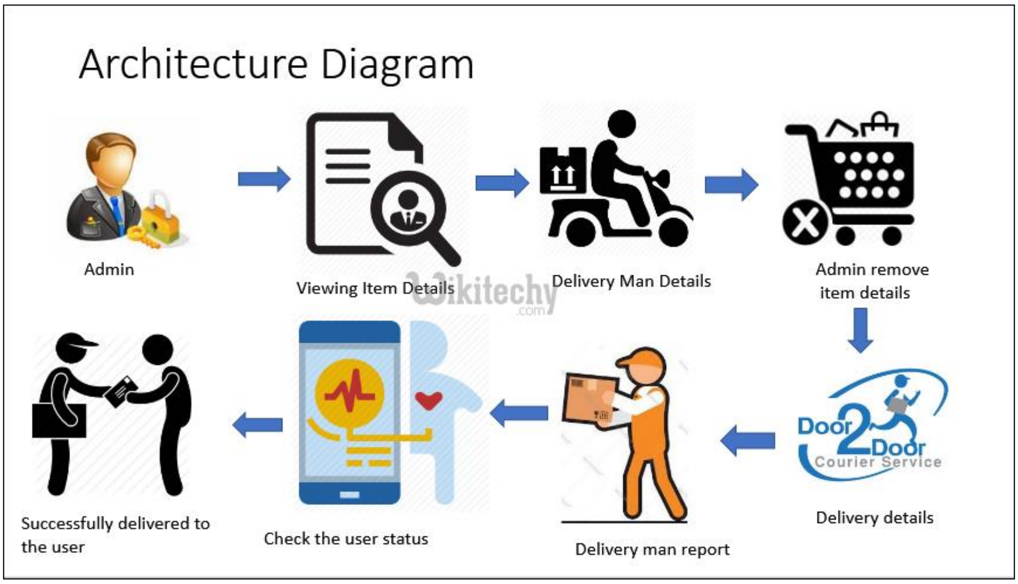
4	**Domain Name and Hosting Cost** • Domain Name Register Cost • Hosting	
		Sub Total
	Grand Total	

Chapter 4

System Design

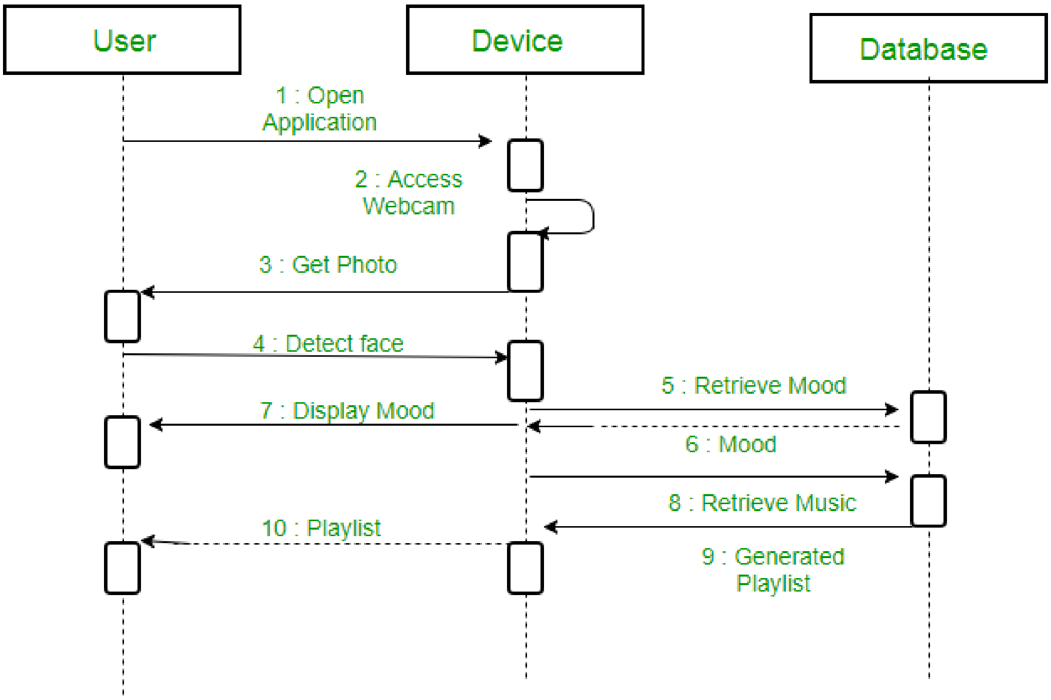
5.1 System Architecture Diagram

< Specify System Architecture Diagram with description>



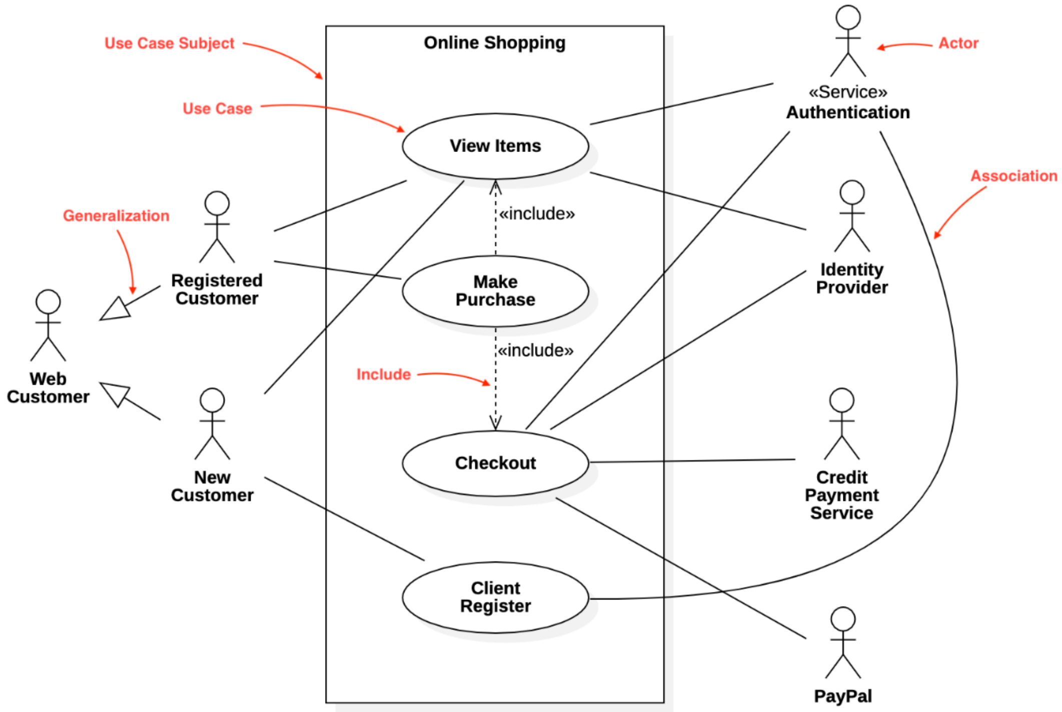
5.2 High Level Design: System Operations

< Specify UML Sequence Diagram with description>



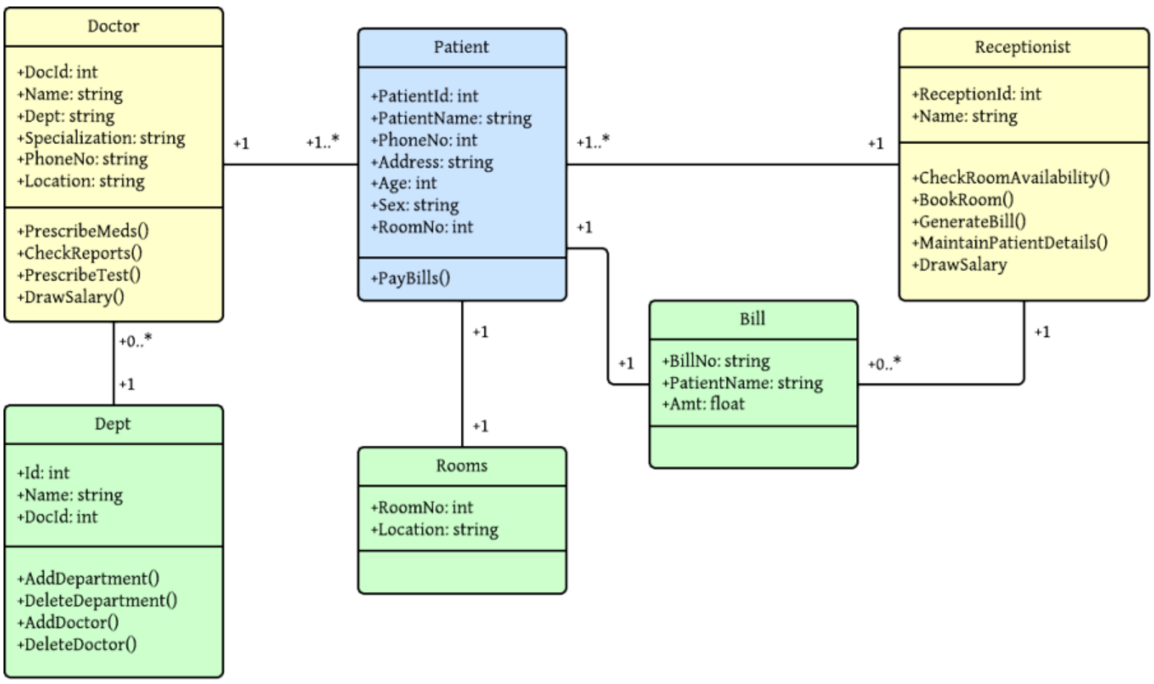
5.3 High Level Design: System Model

<Specify UML UseCase Diagram with description>



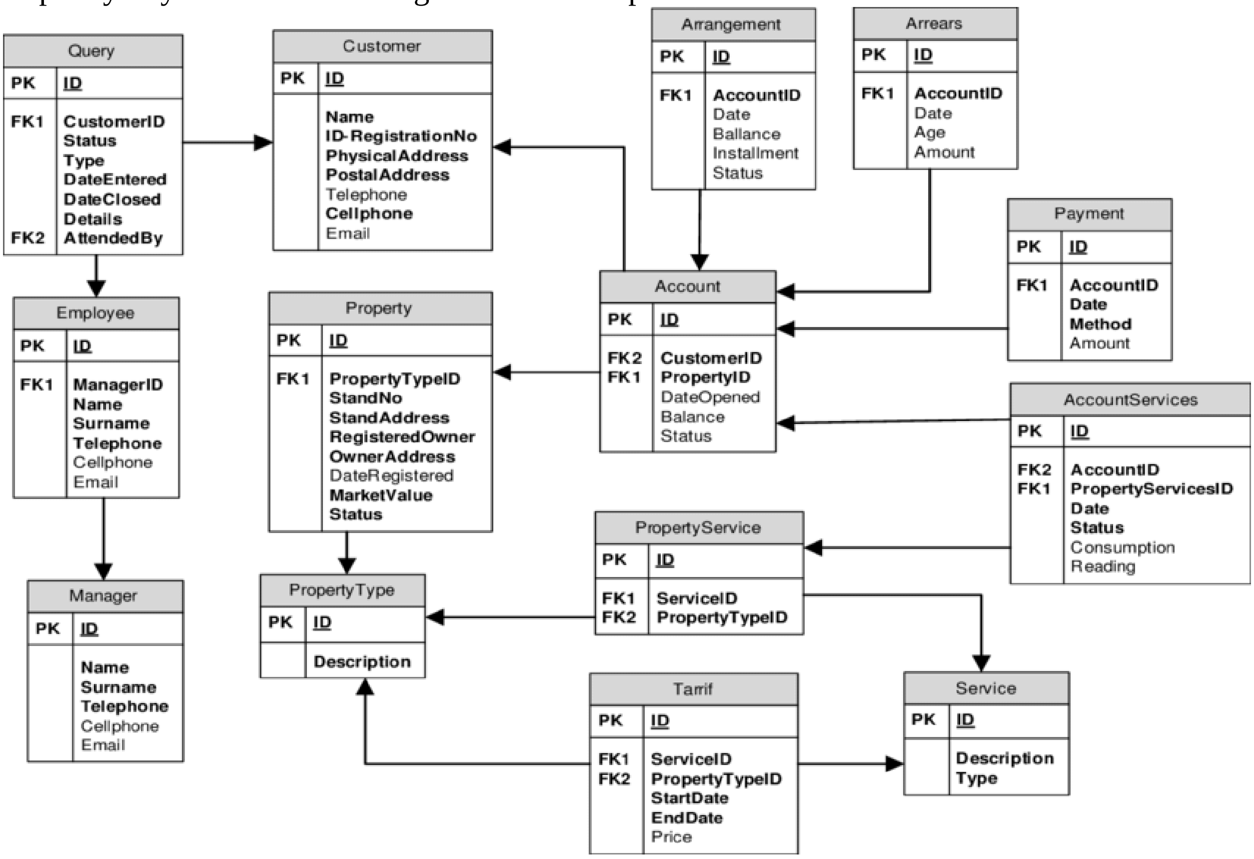
5.4 Low Level Design

<Specify UML Class Diagram with description>



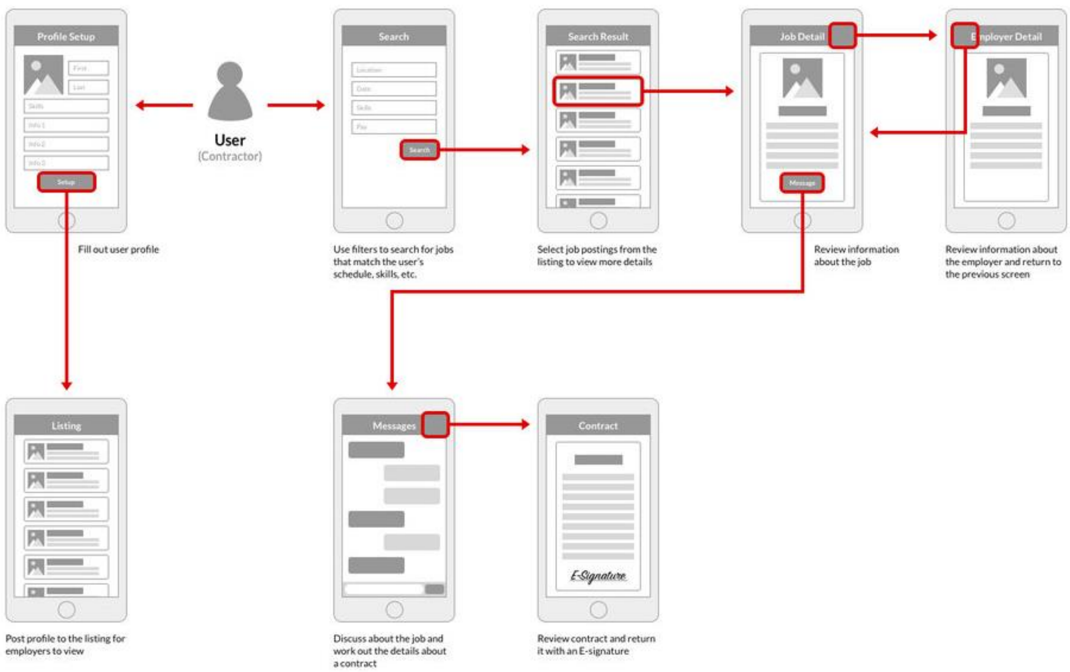
5.5 Database Design

<Specify Physical Database Diagram with description>



5.6 GUI Design

<Show your wireframes with user flow >



Chapter 5

Business Model

5.1 Business Model Canvas

The business model canvas is a tool designers use to map out a business or product's key actors, activities and resources, the value proposition for target customers, customer relationships, channels involved and financial matters. It gives an overview to help identify requirements to deliver the service and more.

5.1.1 Key Partners

Who are our Key Partners? Who are our key suppliers? Which Key Resources are we acquiring from partners? Which Key Activities do partners perform?

MOTIVATIONS FOR PARTNERSHIPS: Optimization and economy, Reduction of risk and uncertainty, Acquisition of particular resources and activities

5.1.2 Key Activities

What Key Activities do our Value Propositions require? Our Distribution Channels? Customer Relationships? Revenue streams?

CATEGORIES:

Production, Problem Solving, Platform/Network

5.1.3 Key Resources

What Key Resources do our Value Propositions require? Our Distribution Channels? Customer Relationships Revenue Streams?

TYPES OF RESOURCES: Physical, Intellectual (brand patents, copyrights, data), Human, Financial

5.1.4 Value Propositions

What value do we deliver to the customer? Which one of our customer's problems are we helping to solve? What bundles of products and services are we offering to each Customer Segment? Which customer needs are we satisfying?

CHARACTERISTICS: Newness, Performance, Customization, “Getting the Job Done”, Design, Brand/Status, Price, Cost Reduction, Risk Reduction, Accessibility, Convenience/Usability

5.1.5 Customer Relationships

What type of relationship does each of our Customer Segments expect us to establish and maintain with them? Which ones have we established? How are they integrated with the rest of our business model? How costly are they?

5.1.6 Channels

Through which Channels do our Customer Segments want to be reached? How are we reaching them now? How are our Channels integrated? Which ones work best? Which ones are most cost-efficient? How are we integrating them with customer routines?

5.1.7 Cost Structure

What are the most important costs inherent in our business model? Which Key Resources are most expensive? Which Key Activities are most expensive?

IS YOUR BUSINESS MORE: Cost Driven (leanest cost structure, low price value proposition, maximum automation, extensive outsourcing), Value Driven (focused on value creation, premium value proposition).

5.1.8 Revenue Streams

For what value are our customers really willing to pay? For what do they currently pay? How are they currently paying? How would they prefer to pay? How much does each Revenue Stream contribute to overall revenues?

TYPES: Asset sale, Usage fee, Subscription Fees, Lending/Renting/Leasing, Licensing, Brokerage fees, Advertising

FIXED PRICING: List Price, Product feature dependent, Customer segment dependent, Volume dependent

DYNAMIC PRICING: Negotiation (bargaining), Yield Management, Real-time-Market

SAMPLE CHARACTERISTICS: Fixed Costs (salaries, rents, utilities), Variable costs, Economies of scale, Economies of scope

References:

[It should be IEEE Format]